

Isha Pali

Research Interests: Network Security, Automotive security, Adversarial machine learning

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Amlidih, Raipur, 492001 (C.G)

Education

Ph.D. in Computer Science and Engineering: University of Queensland – IIT Delhi Research Academy (**UQIDRA**)

Supervisors: [Dr. Dan Kim \(UQ\)](#), [Dr. Vireshwar Kumar \(IITD\)](#), [Dr. Naipeng Dong \(UQ\)](#)

M.Tech in Computer Science Engineering: DSPM IIIT-Naya Raipur | CGPA: 9.4 / 10 | Silver Medalist (Highest CGPA)

Supervisor: [Dr. Ruhul Amin \(IIITNR\)](#)

B.Tech in Computer Science Engineering NIIT University, Rajasthan | CGPA: 9.47 / 10

Minor in Finance (CGPA: 9.67 / 10)

Technical Skills

Core Technologies & AI/ML: Python, C++, P4, PyTorch, scikit-learn, Adversarial Robustness Toolbox (ART)

Systems & Networking: SDN, BMv2, Mininet, Wireshark, CARLA, ProVerif, Scyther

Awards

- Won the **Best Poster award and cash prize of INR 3k** at UQ-IITD Annual Symposium 2026.
Poster Title: “Can Hackers fool a Car’s Digital Immune System?”
 - **GATE qualified** — Score: 418, Rank: 8022
 - **participated in the Grand Finale** — Smart India Hackathon 2018 held in Kolkata.
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Industry & Teaching Experience

TATA motors project | funded automotive security research | IIT Delhi Jan 2025 - present

- Designed and evaluated adversarial attacks against ML-based CAN bus IDS, validated in the CARLA autonomous driving simulator.
- Developing defence mechanisms to improve the robustness of in-vehicle intrusion detection against adversarial attacks.

Academic Tutor | The University of Queensland Jan 2024 – Dec 2024

- Tutored two courses across two semesters for undergraduate and postgraduate students at UQ
 - *CYBR3000 Information Security*
 - *COMS3200 Computer Networks*

Intern (Cyber Security Analyst) | PricewaterhouseCoopers (PwC) — Jan 2019 – July 2019

- Built a Python/Selenium automation pipeline that replaced a manual report generation and delivery workflow, reducing turnaround time significantly.
 - Developed employee security awareness training material aligned to the client's Code of Ethics and Information Security Policies.
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Publications

Journals

- **Pali, I.**, Abdussami, M., Amin, R., Al-Shehari, T., Al-Razgan, M., & Alfakih, T. (2024). S2DN: Design of robust authentication protocol with session key establishment in multi-controller based software-defined VANETs. ***Vehicular Communications***, 47, 100767.
- **Pali, I.**, Amin, R., & Abdussami, M. (2024). Autonomous vehicle security: Current survey and future research challenges. ***Security and Privacy***, 7(3), e367.
- Amin, R., **Pali, I.**, & Sureshkumar, V. (2021). Software-defined network enabled vehicle to vehicle secured data transmission protocol in VANETs. ***Journal of Information Security and Applications***, 58, 102729.

Conference

- **Pali, I.**, Thakur, S., Maurya, V., Dong, N., Kumar, V., & Kim, D. D. Poster: Mind the Structural Gap-Evaluating Automotive CAN Datasets as Interchangeable Benchmarks for Intrusion Detection, **Usenix Vehiclesec'26**
- **Pali, I.**, & Amin, R. (2022, May). PortSec: Securing port knocking system using sequence mechanism in SDN environment. In 2022 ***International Wireless Communications and Mobile Computing (IWCMC)*** (pp. 1009-1014). IEEE. (core B)

Preprint / Under review

- **Pali, I.**, Krishania, L., Chadha, D., Kandar, A., Varshney, G., & Shukla, S. (2020). A Comprehensive Survey of Aadhar and Security Issues. *arXiv:2007.09409*.
- PLAID: Practical Machine Learning-Assisted Robustness Framework against Automotive Intrusion Detection. Submitted to **ASIA CCS 2027** (CORE A), currently under review.

Talks

- **Guest for an Invited talk** at **Oracle Labs Brisbane**, Australia on **14 December 2024**.
- Presented a talk at **UQ Cyber Security Research Conference (CSRC 2024)**, **11 July 2024**.
- Presented my research at **UQ – IITD Knowledge Nexus Seminar Series** - India-Australia Energy Corridor.